

AI BlogNest API

AI-PoweredBlogManagementand Content Generation REST API Platform

PHASE 2: REQUIREMENT ANALYSIS

Role	Team Member
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Submitted as a phase-wise project submission.

1. Functional Requirements

- **User Authentication & Session Management:** Support secure user registration, credential verification, password hashing with bcrypt.js, and stateless JWT token issuance.
- **Role-Based Access Control (RBAC):** Grant endpoint access according to user privileges (Admin, Editor, Author, Reader) to guard sensitive actions.
- **Blog Lifecycle Management:** Enable complete CRUD (Create, Read, Update, Delete) execution on blog posts, supporting status states such as Draft, Scheduled, and Published.
- **AI Content Generation Service:** Accept topic and category inputs to automatically synthesize full-length blog posts via Google Gemini API SDK.
- **AI Content Summarization Service:** Accept body text payloads and return concise, context-aware summaries.
- **Category & Comment Management:** Enable taxonomic organization via categories/tags and provide feedback endpoints with administrative comment moderation.

2. Non-Functional Requirements

- **Security & Input Sanitization:** Prevent XSS, parameter pollution, and NoSQL injection vulnerabilities through strict schema validation and middleware filtering.
- **Scalability & Maintainability:** Structured around a decoupled Model-View-Controller (MVC) pattern for modular expansions and clean service segregation.
- **Performance & Efficiency:** Low latency API response times for standard CRUD operations and fast JSON payload parsing.
- **Reliability & Logging:** Centralized error handling middleware and environment-based request logging via morgan

3. Technology Requirements

Technology / Tool	Purpose & Usage Description
Node.js (v16+)	Asynchronous, event-driven JavaScript runtime ecosystem hosting the server infrastructure.
Express.js	Lightweight web framework utilized for RESTful routing, middleware composition, and controller binding.
MongoDB & Mongoose	NoSQL document database paired with Mongoose ODM for strict schema modeling and query execution.
Google Gemini AI SDK	Generative AI library integration powering automated blog generation and text summarization services.
JWT & Bcrypt.js	JSON Web Tokens for stateless identity verification and bcrypt.js for secure multi-round password hashing.
Thunder Client / Postman	API verification toolkits used for testing REST routes, token authorization headers, and response payloads.

4. Environment Configuration

- **Operating System:** Windows 10/11, macOS, or Linux (cross-platform Node.js runtime).
- **Hardware Specifications:** Intel Core i5 (8th Gen+) or AMD Ryzen 5 processor, 8 GB RAM minimum (16 GB recommended), and at least 1 GB disk workspace.
- **Software Prerequisites:** Node.js installed with npm package manager and Visual Studio Code IDE.